REFLECTION AND REFRACTION (SNELL'S LAW)

1. Reflection of Light

Definition: Reflection is the phenomenon in which light rays bounce back into the same medium after striking a surface.

Laws of Reflection

1. The incident ray, reflected ray, and normal at the point of incidence all lie in the same plane.
2. The angle of incidence is equal to the angle of reflection.

Formula:

$$\boxed{i=r}$$

Where,

- i = Angle of incidence
- r = Angle of reflection

Types of Reflection

1. Regular Reflection: Occurs when light falls on a smooth, polished surface. The reflected rays remain parallel.
2. Diffuse Reflection: Occurs when light falls on a rough surface. The reflected rays scatter in different directions.

Applications of Reflection

1. Mirrors and periscopes.
2. Reflecting telescopes.
3. Vehicle rear-view mirrors.
4. Reflectors used in headlights.

2. Refraction of Light

Definition: Refraction is the change in direction of light when it passes from one transparent medium to another due to a change in its speed.

Laws of Refraction

1. The incident ray, refracted ray, and normal at the point of incidence lie in the same plane.
2. For a given pair of media and a given wavelength, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.

3. Snell's Law

Snell's law describes the relationship between the angles of incidence and refraction and the refractive indices of two media.

Formula:

$$\boxed{n_1\sin i=n_2\sin r}$$

Where,

- n_1 = Refractive index of the first medium
- n_2 = Refractive index of the second medium
- i = Angle of incidence
- r = Angle of refraction

For light travelling from air into a material, the refractive index of the material is approximately:

$$\boxed{n=\frac{\sin i}{\sin r}}$$

4. Refractive Index

The refractive index of a medium is the ratio of the speed of light in vacuum to its speed in that medium.

Formula:

$$\boxed{n=\frac{c}{v}}$$

Where,

- n = Absolute refractive index
- c = Speed of light in vacuum
- v = Speed of light in the medium

The refractive index is dimensionless.

5. Bending of Light During Refraction

1. Rarer to Denser Medium: Light bends towards the normal because its speed decreases.
2. Denser to Rarer Medium: Light bends away from the normal because its speed increases.
3. Along the Normal: When light enters a different medium perpendicularly, it does not change direction, although its speed changes.

6. Difference Between Reflection and Refraction

Reflection	**Refraction**
Light returns to the same medium.	Light enters another medium.
Light changes direction at a surface.	Light generally changes direction due to a change in speed.
Angle of incidence equals angle of reflection.	Governed by Snell's law.

The reflected ray remains in the original medium.	The refracted ray travels in the second medium.
Example: Plane mirror.	Example: Light passing from air into glass.

7. Applications of Refraction

1. Lenses used in spectacles.
2. Cameras and microscopes.
3. Telescopes and binoculars.
4. Optical instruments.
5. Optical fibres, which guide light using total internal reflection.

8. Conclusion

Reflection is the bouncing back of light from a surface, whereas refraction is the change in direction of light when it passes between media with different refractive indices. Snell's law gives the mathematical relationship between the angles of incidence and refraction.